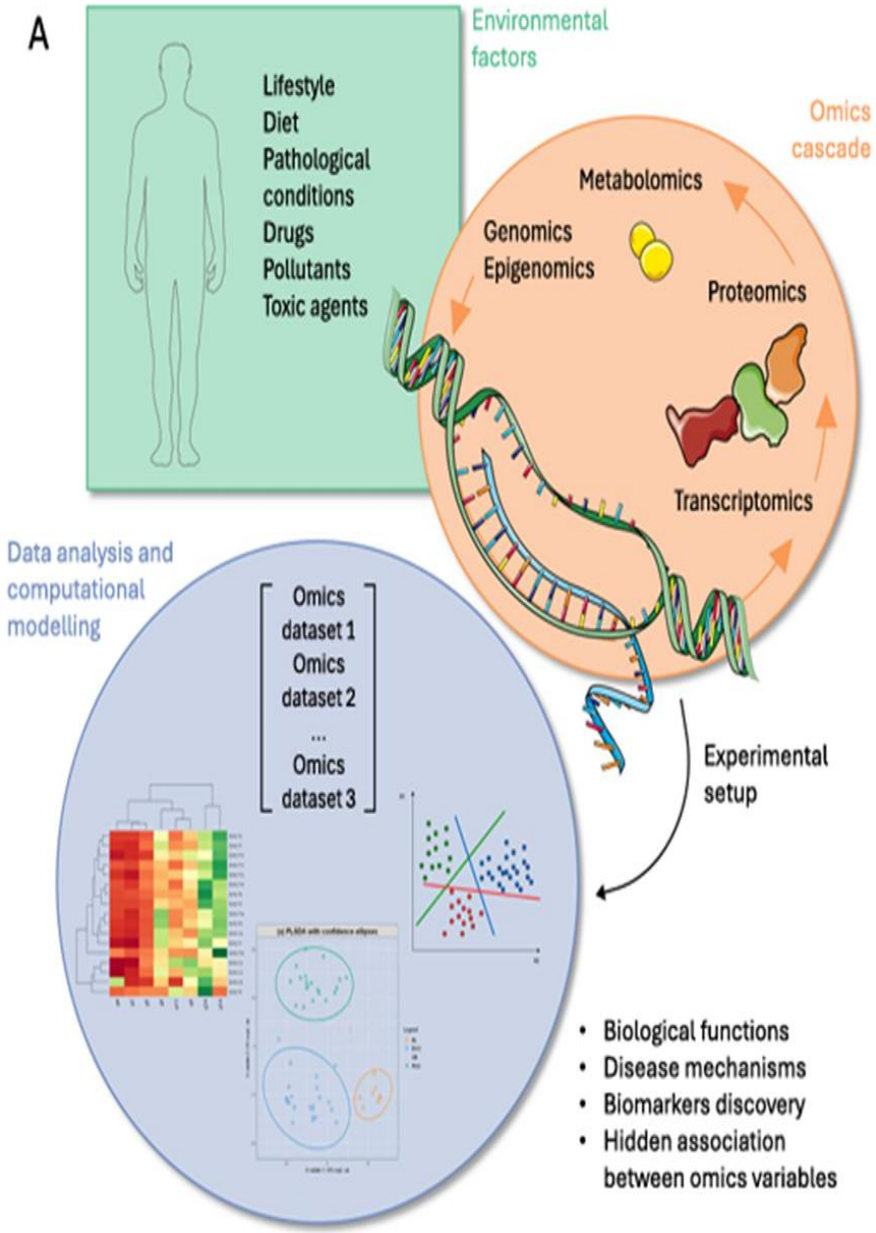
Algorithms And Tools For Data-driven Omics Integration To Achieve Multilayer Biological Insights

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The Importance of Multi-Omics Integration in Systems Biology

•**Systems biology** integrates experimental and computational methods to study biological processes at multiple scales.

•**Omics technologies** (genomics, transcriptomics, proteomics, metabolomics) generate massive datasets describing molecular layers.

•Integration of these datasets enables **multilayer understanding** of diseases and **biomarker discovery**.

Despite its promise, **data integration is challenging** due to:
➢High data dimensionality and heterogeneity
➢Missing values and variable quality
➢Complexity of multi-layer interactions

Multi-omics integration lies at the core of systems biology, offering deeper insights into health and disease — but it demands advanced computational and statistical strategies.

- Integration methods vary in complexity, interpretability, and biological depth
- Each method addresses different challenges: dimensionality, heterogeneity, correlation, and classification
- Review includes 64 studies (2018–2024) using data-driven omics integration.

Statistical and Correlation-Based Methods

Correlation analysis is the foundation for integrating omics data and uncovering molecular relationships.

- **Correlation quantifies relationships** between variables across omics layers.
- Simple visualization (e.g., **scatterplots**) helps reveal consistent or divergent gene–protein or transcript–metabolite patterns.
- **Pearson's and Spearman's correlation** are most common:
  - Pearson: linear relationships
  - Spearman: rank-based, non-linear associations
- Used to assess transcription–protein correspondence, identify regulatory pathways, and detect time delays between mRNA and protein production.

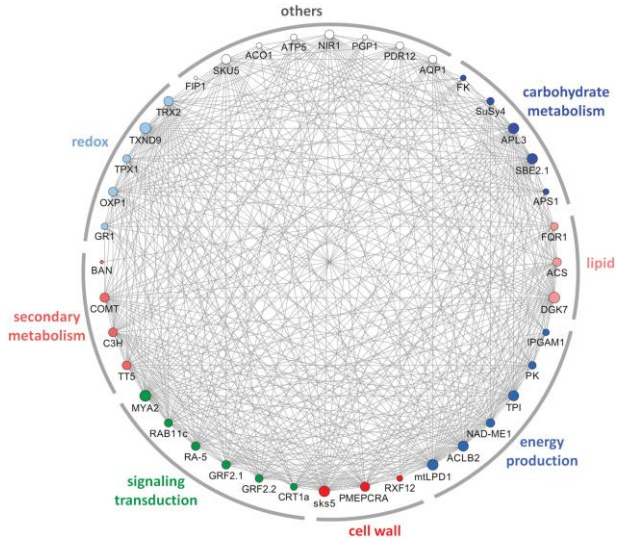
Example Studies:

- **Zheng et al.**: color-coded scatterplots for transcription efficiency.
- **Gao et al.**: transcript-to-protein ratio patterns.

Advanced Correlation Measures and Network Extensions

Beyond pairwise analysis, correlations can be generalized to explore complex multi-omics relationships.

- **RV coefficient** — multivariate extension of Pearson correlation, compares whole datasets.
- **Procrustes analysis** — aligns datasets geometrically to measure similarity.
- **Correlation networks:**
  - Nodes = genes/proteins/metabolites
  - Edges = significant correlations (based on R^2 or p-values)
  - Enables visualization of **interconnected molecular systems**.
- Example: **Gong et al.** built a multi-omics co-expression network integrated with cancer pathways.



The study constructed a co-expression network showing that genes expressed alongside key starch biosynthesis genes (APL3, APS1, SBE2.1) in cassava roots are mainly involved in carbohydrate, energy, and related metabolic pathways throughout root development. Ding Z a et al. 2020

Weighted Gene Correlation Network Analysis (WGCNA) and xMWAS

Network-based correlation approaches reveal functional modules and community structures.

- **WGCNA** identifies **modules of highly correlated genes** by building a **scale-free network**.
  - Modules summarized by **eigenmodules** linked to clinical traits.
  - Used to connect transcript/protein and metabolite clusters (e.g., *Ding et al.*).
- **xMWAS:**
  - xMWAS performing **correlation and multi-omics integration analysis**.
  - Combines correlation and multivariate PLS components.
  - Builds **multi-omics integrative networks** and detects **communities** of related features using modularity optimization.
  - Applied to uncover hidden **biological pathways** and **disease mechanisms** (e.g., *Na et al.*).

xMWAS:

It takes omics datasets (such as transcriptomics, metabolomics, proteomics, etc.) **organized in matrices and performs pairwise association analysis** between features from different omic layers.

The core of the method relies on **Partial Least Squares (PLS) regression** — a statistical technique that projects high-dimensional data into a lower-dimensional space by finding **latent components** that **maximize the covariance between two datasets.** In other words, PLS identifies combinations of variables (components) from each dataset that are most strongly correlated, making it particularly suitable for multi-omics data where the number of variables is large and they may be collinear.

Using PLS components and regression coefficients, xMWAS calculates the correlation strength between omic features and constructs an **integrative association network**, where nodes represent biological entities (e.g., genes, metabolites, proteins) and edges represent **statistically significant associations.**

Next, xMWAS applies the **multilevel community detection algorithm** to identify **clusters or communities** of highly connected nodes. **This algorithm iteratively maximizes a measure called modularity, which quantifies how strongly nodes within a cluster are connected compared to those outside it**.

Finally, xMWAS uses these communities to uncover **biological pathways** and **interconnections across omic layers**, revealing how molecular features from different omics interact functionally.

Canonical Correlation Analysis (CCA) and its high-dimensional variant, **Regularized Canonical Correlation Analysis (rCCA):**

are multivariate statistical methods designed for **integrative analysis and dimensionality reduction** across **two omics datasets**. Both methods aim to identify and quantify the **correlations between two sets of variables** (e.g., genes and metabolites).

CCA works by finding **canonical variates**, which are **linear combinations of variables** from each dataset that maximize their mutual correlation. Each pair of canonical variates corresponds to a **canonical correlation coefficient**, representing how strongly the two derived components are associated.

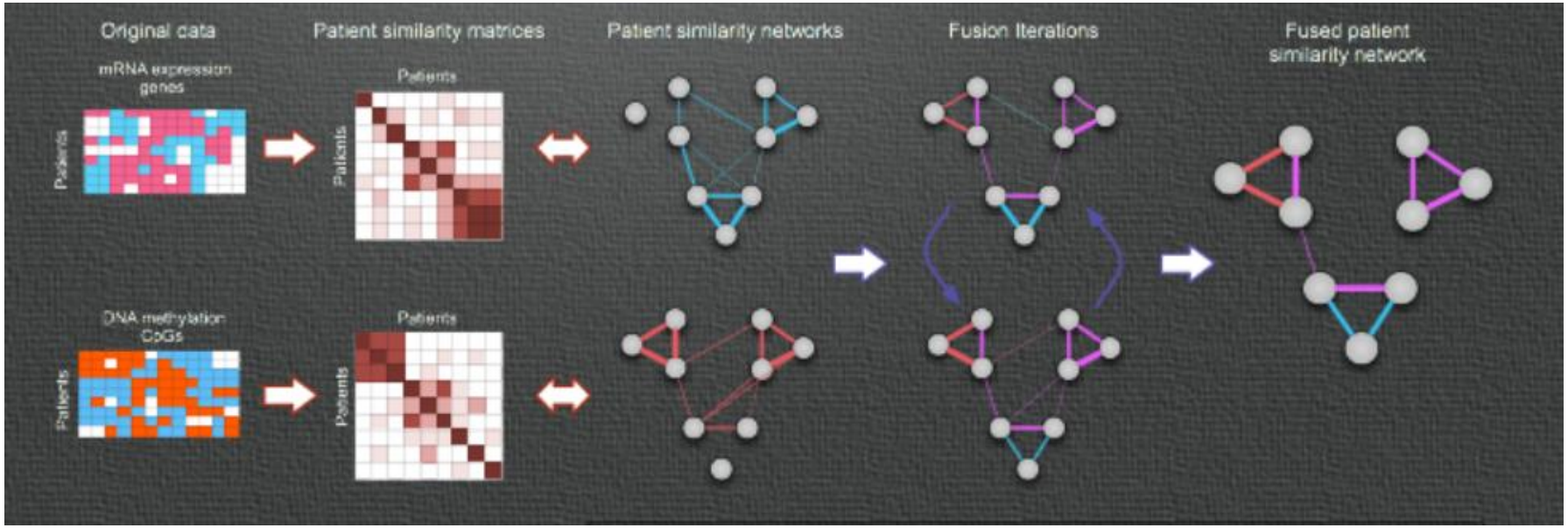
However, standard CCA performs poorly when the number of variables greatly exceeds the number of samples or when features are highly collinear — a common issue in omics data.

To address this, **rCCA (Regularized CCA)** introduces an **L2 (Ridge) penalty** to the diagonal of the covariance matrices. This **regularization** stabilizes the matrix inversion process and reduces overfitting, making the method suitable for **high-dimensional, noisy biological datasets**.

Using rCCA, researchers can also construct **relevance networks**, where nodes represent variables from different omics datasets, and edges connect variables with strong canonical correlations. These networks visualize **cross-omic relationships** and **help identify key features (highly connected nodes) t**hat may play central biological roles, such **as biomarkers or regulators involved in disease mechanisms.**

Similarity Network Fusion (SNF)

- **Similarity Network Fusion (SNF):**
  - Constructs **sample similarity networks** for each omic dataset.
  - Iteratively fuses them using **graph-based message-passing**, enhancing shared structure.
  - Enables **clustering of patients** and **disease subtype discovery**.
  - Shown to outperform single-omics in disease classification (*Ruan et al., scSNF* variant).



Multivariate Methods for Multi-Omics Integration

- Represent the **most extensive and diverse class** of integration strategies.
- Rely on **algebraic decompositions** and **latent variables** to extract shared patterns.
- **Latent variables** = inferred algebraic coordinates representing shared data structures
- → reduce dimensionality & highlight relationships.
- Enable detection of **common biological signals** across omics layers.

Principal Component Analysis (PCA) and Extensions

PCA: These components are uncorrelated and serve as simplified representations of the original features, allowing easier interpretation and **reducing redundancy capturing maximum variance.**

SUM-PCA is an extension of PCA for **multi-block or multi-omics data**, where multiple data types are measured on the same samples.

In this method, all datasets are combined into a single matrix, and PCA is applied to extract **super scores** that summarize shared variation across all data blocks.

Each block retains its own **loadings** and **residuals**, while the super scores represent a **consensus pattern** among datasets.

This approach provides an initial overview of how different omics layers capture similar biological behaviors.

Multi-Omics Factor Analysis (MOFA)

Is a data-driven method that extends PCA for multi-omics integration.
Instead of finding components that capture overall variance like PCA, MOFA identifies a set **of hidden (latent) factors** that explain the **main sources of variability across multiple omics datasets** — such as transcriptomics, proteomics, or metabolomics.

Each omic data matrix is decomposed into two parts:
a latent factor matrix shared across all omics, and a **weight matrix specific to each omic type**, plus a residual noise term. This structure allows MOFA to distinguish between shared sources of variation (common across omics) and unique sources (specific to one omic).

MOFA+: Scalable for single-cell datasets.
Handles millions of cells; integrates cell structure info.

MEFISTO: Adds temporal/spatial dimensions → e.g., COVID-19 proteomic dynamics.

Multiple Factor Analysis (MFA)

is an extension of PCA designed for **analyzing multiple groups of variables**, such as different omics datasets or data blocks.

The key idea is that each group of variables (for example, transcriptomics, proteomics, or metabolomics) is first analyzed separately using PCA.

Then, MFA assigns a **weight** to each group — specifically, the inverse of the first eigenvalue from its individual PCA — so that no group dominates the overall analysis due to scale or dimensionality differences.

By performing a global PCA on these **weighted variables**, MFA ensures that all data blocks contribute equally to the integrated analysis.
The resulting **factorial axes** (or dimensions) help identify patterns, similarities, and discrepancies between data blocks, showing which omics datasets provide consistent or distinct information about the samples.

MFA is particularly useful for **pattern recognition and data integration**, offering a **balanced and interpretable view across multiple data types.**

DIABLO and sGCCA

It is a **supervised** multi-omics integration tool designed to identify robust biomarkers **using Latent variable approaches** for Omics studies.

The method extends sparse Generalized Canonical Correlation Analysis (sGCCA) a multivariate dimension reduction approach based on Singular Value Decomposition (SVD) .SVD de—composes a data matrix into components that **capture the most relevant variation**, allowing DIABLO to project data **from multiple omics layers into a shared low-dimensional space, called latent components.**

Unlike unsupervised methods, DIABLO incorporates **class labels** (through a dummy matrix) to find components that not only maximize correlations across omics datasets but also best discriminate between sample groups (e.g., healthy vs. diseased).

It uses **L1 penalization (LASSO)** to perform variable selection, meaning it identifies only the most informative features from each dataset.

Partial Least Squares (PLS)-based methods

projection-based multivariate techniques that explore the relationship between two or more continuous variables. **PLS projects data onto latent components that maximize the covariance between predictors (X) and responses (Y).**

While originally designed for regression, its variant **PLS-Discriminant Analysis (PLS-DA)** handles **categorical response variables**, focusing on maximizing class separation while capturing variance in X.

Sparse versions of PLS and PLS-DA apply an **L1 penalty** to select **relevant variables** in high-dimensional omics data.

One key feature of PLS-DA is the **Variable Importance in Projection (VIP)** score, which measures each **variable's contribution to explaining the variance in X and Y**.

The **Backward Elimination PLS-DA (BE PLS-DA)** method further refines the model by **iteratively removing variables with the lowest VIP scores**, improving class discrimination and model simplicity.

Co-Inertia Analysis (CIA) and Multiple Co-Inertia Analysis (MCIA)

CIA and MCIA are multivariate methods designed to explore **relationships and shared structures across multiple datasets**, and are widely used for **integrative omics analysis**.

CIA, performs **simultaneous ordination of two datasets** to identify **successive orthogonal axes** that maximize the **squared covariance** between them. This allows visualization of **joint similarities and co-variation patterns**. For instance, CIA has been applied to detect **co-expression between proteins and genes** in brain ischemia.

MCIA extends CIA to **more than two datasets**. It projects each omics dataset into a **common low-dimensional space**, **maximizing the sum of squared covariances** among all datasets. In this shared space, features or samples with similar patterns are located close together, revealing **global correlations and consistencies** across omics layers.

Machine Learning and Artificial Intelligence Approaches

•ML enables systems to analyze **complex omics data**, detect **patterns**, and make **predictions or classifications** automatically.

•Two main paradigms:
•**Supervised learning:** model trained on labeled data → predicts known outcomes.
•**Unsupervised learning:** explores unlabeled data → identifies hidden structures or patterns.

•Examples:
•**Supervised:** linear/logistic regression, decision trees, random forests, SVMs, neural networks.
•**Unsupervised:** clustering (k-means, hierarchical), dimensionality reduction (PCA, AE).

•Increasingly important for **multi-omics integration** due to data complexity and non-linear relationships.

Unsupervised Learning in Multi-Omics Integration: Clustering-Based Approaches

Aim: group samples based on omics similarity without prior labels.

✓**Consensus Clustering (ConsensusClusterPlus)**

✓**Integrative Clustering (iClusterBayes)**

✓**Subspace Clustering (MineClus)**

Integrative Clustering (iClusterBayes)

In this methods, data are clustered in a **latent variable space**. The iClusterBayes package implements a Bayesian latent variable model to **jointly analyze multiple omics datasets from the same samples.**

Integrative clustering has mainly been used to identify **patient subgroups with distinct molecular and clinical profiles**.

ConsensusClusterPlus

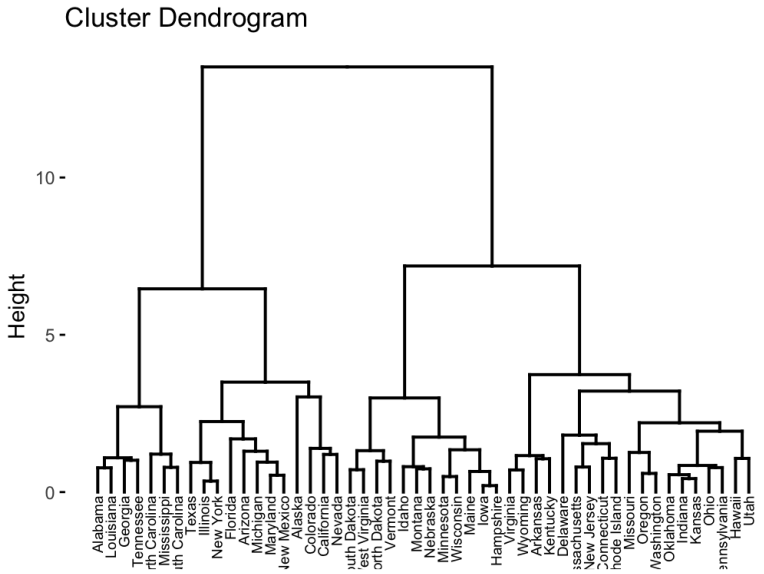
Consensus clustering method provides a more robust approach to clustering **by combining results from multiple iterations of clustering.**

ConsensusClusterPlus: a unsupervised method repeatedly partitions subsets of samples and features into k groups using a clustering algorithm.

The pairwise consensus value—the proportion of times two items cluster together—is computed for all item pairs and stored in a consensus matrix.

Using these values, an **agglomerative hierarchical consensus** clustering is built, where the distance is defined as 1 − consensus value, and clusters are pruned into k consensus groups.

This approach has been applied to **large-scale cancer subtype discovery**. For instance, Liu et al. used it to **identify four molecular subtypes of esophageal squamous cell carcinoma,** each linked to specific biomarkers and therapeutic targets.



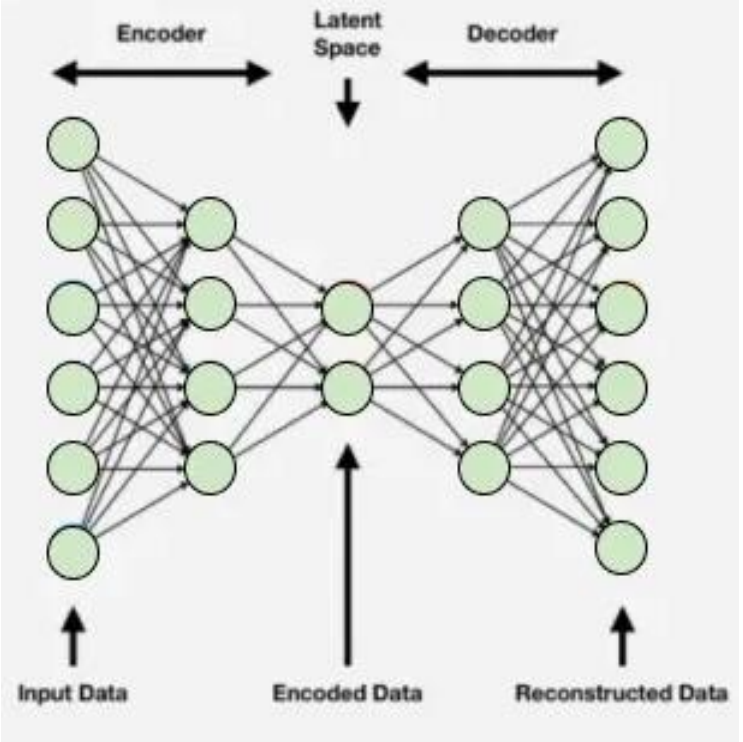
Subspace Clustering (MineClus)

Subspace clustering is a recent clustering-based strategy designed to detect clusters within **specific subspaces** of **high-dimensional multi-omics data**, aims to **uncover local structures that may not be visible in the full-dimensional space.**

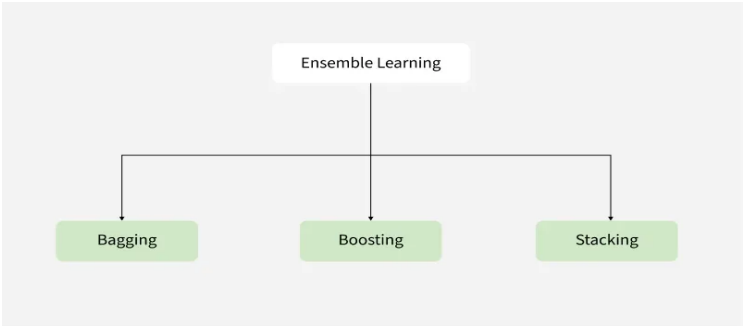
In this framework, autoencoders (AEs) — **a class of deep neural networks — are first used to reduce the dimensionality o**f proteomic, transcriptomic, and metabolomic data.

AEs compress the input data into a **low-dimensional latent representation** by **learning to reconstruct the original input with minimal error, optimizing the weights of their activation functions through training.**

After obtaining the AE embeddings, all omics layers **are horizontally concatenated to form a unified representation** for subspace clustering.

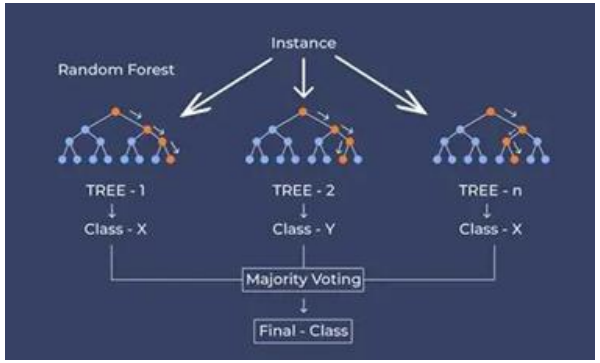


Supervised Learning for Prediction and Classification:



Regression Models:
Capture linear relationships between omics features and outcomes.
Once fitted, the regression line can be used to predict the outcome value of new input data.
Elastic-Net Regression combines LASSO and ridge → avoids overfitting.

Ensemble learning:
We use many small models instead of just one.
Each of these models may not be very strong on its own, but when we put their results together, we get a better and more accurate answer.



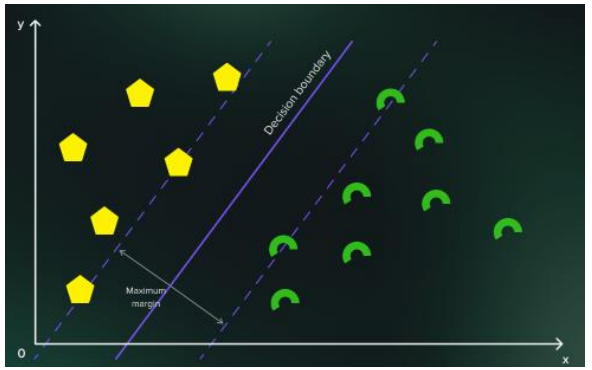
- **Random Forest (RF):** Used for multi-omics classification.
- **AdaBoost & Gradient Boosting Machines:** iterative improvement; applied to multi-omics disease response prediction.

AdaBoost :combines decision trees but it assigns an equal weight to the training data, **then it calculates the errors** and **increases the weight of misclassified data points**. Another tree is then fitted on the same datas**et with updated weights,** and the process is repeated iteratively until all the trees are fitted.

Gradient Boosting Machines : This algorithm works by constructing **additive regression models iteratively minimizing residuals** by least squares.

- **Support Vector Machines (SVM):** identify optimal separating hyperplanes between disease classes.

- **Deep Learning (Neural Networks):** model nonlinear patterns across omics layers; used for **cancer diagnosis** and **phenotype prediction**.



- **AutoGluon-Tabular:**

is an **automated ML algorithm** designed to **build predictive models** from unprocessed tabular datasets, such as CSV files. **It simplifies the modeling process by automatically** recognizing data types in each column, including text, and optimizing **hyperparameters and feature engineering**.

Automated ML framework **combining multiple base models** (RF, LightGBM, CatBoost, XGBoost, neural nets). These base models are trained, and **their predictions are used as features to train a final ensemble model**, **which combines the strengths of the base models.**

Challenges and Future Directions in Omics Integration

Integrating diverse omics layers enables **deeper biological insights** but introduces multiple **analytical and computational challenges**.

Key challenges stem from:

- Data **quality, heterogeneity, and imbalance**.
- **Missing data** and **collinearity** across molecular layers.
- **High dimensionality** relative to sample size (p >> n).

- Quality control is crucial:

"**Garbage in, garbage out**" → analysis reliability depends on **robust, reproducible input data**.
Outliers can distort downstream analyses.

- Omics datasets differ in **scale, units, noise, and feature count** (e.g., thousands of genes vs. hundreds of proteins), complicating integration.

Handling Missing Values in High-Throughput Omics

Missing data is common in proteomics & metabolomics.

Classified as:
•**MCAR (Missing Completely at Random)** → independent of data.
•**MAR (Missing at Random)** → depends on other variables.
•**MNAR (Missing Not at Random)** → depends on the missing value itself.

Solutions:
•**Univariate Imputation:** replace missing values by mean, median, mode, or constant values.
•**Multivariate Imputation:** uses relationships among features (e.g., **KNN imputation**).
•**Probabilistic minimum (LCMD)** — replaces missing values with the lowest detectable intensity.

Limitations:
•Imputation may distort biological meaning if absence reflects true lack of molecule.
•ML models like **Random Forests** can sometimes handle missing data directly.

Coping with the Curse of Dimensionality

Omics data often have **few samples (n)** and **many features (p)** → overfitting risk.
•**Rule of thumb:** $n/p \geq 5$ for reliable statistical power.

•**Simple filtering methods:**

•Keep features present in ≥ threshold samples.
•Remove near-zero variance or statistically insignificant variables.
•Retain features correlated with clinical traits.

•**Advanced dimensionality reduction:**

•**PLS, PLS-DA, OPLS-DA, LDA, mRMR** — select discriminative, non-redundant features.
•**Recursive Feature Elimination (RFE, RFECV)** — iteratively removes least-informative features.
•**Machine Learning models** (e.g., Random Forest, Autoencoders) can perform **embedded feature selection**.

•**Risk:** Some models (e.g., PLS-DA) may **overfit or find false projections** if not validated properly.

Multicollinearity Among Omics Variables

- Omics features are biologically interconnected → often **highly correlated (collinear)**.
- **Consequences:** instability in regression coefficients and reduced model interpretability.
- **Detection Methods:**
  - **Variance Inflation Factor (VIF):** measures how much variance increases due to collinearity.
  - **Condition Index (CI)** and **Variance Decomposition Proportion (VDP)** from SVD analysis.
- **Mitigation Strategies:**
  - Remove or combine highly correlated variables (e.g., keep one if $R^2 > 0.9$).
  - Apply **PCA** or other linear combination techniques.
  - Use regularized models (e.g., **Elastic-Net regression**) that tolerate correlation.
- **Caution:** Removing features may eliminate biologically relevant relationships, affecting pathway enrichment analysis.

- **Model Interpretability**
- Complex multivariate and machine learning models **often transform input features into new dimensions**, which may obscure the biochemical meaning of individual variables.
- **Linear models** are inherently interpretable but may sacrifice predictive accuracy.
- Interpretability is crucial for understanding biological relationships and trusting model predictions.
- Balancing complexity and interpretability is essential for reliable biological insights.

Explainable AI (XAI) Approaches

- **LIME (Local Interpretable Model-Agnostic Explanations):** Explains predictions of black-box models by approximating them with simpler, local interpretable models.
- **SHAP (SHapley Additive exPlanations):** Derived from game theory; assigns importance values to each feature contributing to a prediction.

Computational Power and Challenges

High-throughput omics data demand massive computational resources and efficient storage.

Deep learning, Bayesian networks, and graphical models are computationally intensive.

Researchers face challenges in:
-**Processing large datasets** efficiently
-Managing computational **cost**.
-Building scalable **infrastructure.**

Solutions:

Optimization algorithms

Parallel computing

Online learning

Cloud-based computation

Trade-offs between cost and analytical depth remain critical.

Future Directions in Omics Integration

- **Emerging trends aim to overcome data complexity and scalability limitations.**

Single-cell omics:
Captures cellular **heterogeneity and unique cellular functions**.

Spatial omics:
Maps gene expression in tissue context, revealing spatial relationships between cells.

These approaches enhance understanding of **tissue architecture and intercellular signaling.**

- **Machine Learning & Feature Selection in Multi-Omics**

ML and AI are becoming central to omics data integration. Advantages: Handle **high dimensionality, noise, and heterogeneity.**

Feature selection is vital for reducing redundancy and improving model robustness.

Recent methods: Deep learning–based selection, Multi-objective optimization, Quantum computing–driven feature selection — a novel frontier.

Conclusions

- **Omics integration** has emerged as a cornerstone of **systems biology**, enabling the exploration of pathophysiological mechanisms across multiple biological layers.
- It provides complementary insights that **genotypes and phenotypes alone cannot explain**, enhancing diagnosis and prognosis in complex diseases.

➢Multi-omics approaches are key drivers of **personalized medicine**, combining molecular and clinical profiles to enable **tailored healthcare strategies.**

Challenges and Considerations

- Relationships between omics layers are often **associative, not causal** — requiring careful interpretation.
- **Multivariate and AI/ML methods** can uncover hidden structures but may reduce interpretability.
- Success depends on **high-quality, well-designed datasets** and **public data sharing** for reproducibility and collaboration.
- Most **current databases** remain **single-omics focused**, limiting integrative potential.

Thank you for your attention